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Learning to extract buildings from ultra-high-resolution drone images and noisy labels

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ABSTRACT

Building maps have a plethora of applications in government, industry and academia. In most cases, large scale maps can be retrieved from OpenStreetMap vector data. However, for certain rapidly changing built and semi-built environments, corresponding maps are not as accurate and contain label noise such as missing, incorrectly present, shifted labels, etc.; mainly because buildings in those regions are constantly being constructed, deconstructed, replaced and altered. One such case is extant in the Rohingya camps of southeastern border region of Bangladesh. Mass refugee influx in late 2017 and following population growth has necessitated the construction of buildings and expansion of camps. Consequently, reliable methods are necessary for detecting and documenting camp buildings. Ultra-high-resolution drone images of Rohingya camps are semantically segmented through fully convolutional U-Net deep learning systems for generating accurate building maps from noisy labels. A wide variety of noises are prevalent in the labels. Deep learning systems provide less noisy predictions compared to the classification tool in the most widely used Geographic Information System (GIS) software, ArcGIS. Data augmentation and regularization allows reliable learning, even in the presence of label noise. During testing, calculation of numeric performance metrics against noisy labels can grossly underestimate true skill and performance of the model. A subset of 22 million pixels of the testing data is relabelled by hand to obtain noise-free labels. Testing our generated maps against noisy and noise-free labels confirms that true performance is higher than otherwise indicated by freely available building maps. Empirical results reveal that utilized pipeline is able to learn from noisy data and produce labels which are more accurate and less noisy. Labels generated by our best performing system provide Intersection-over-Union (IoU) gain of 17.6% and Dice score gain of 13.6% over freely available labels from OpenStreetMap. Finally, spatio-temporal building maps are generated to portray the applicability of this research.

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1. Introduction

Anthropogenic activities are shaping the Earth's built environment at an unprecedentedly rapid pace. One such global issue in the twenty-first century is forced migration. The largest stateless population in the world, the Rohingya people, also known as Forcibly Displaced Myanmar Nationals (FDMNs), have forcefully migrated to the southeastern border region of Bangladesh in late 2017. In the subsequent years, the refugee population has been increasing, impelling expansion of refugee camps through construction of new buildings which are necessary to accommodate the growing refugee population. Existing studies employing machine learning models and remotely sensed data show that refugee camp area is increasing and forest cover area is decreasing (Hassan et al. 2018; Ahmed et al. 2019a). This region is prone to landslide and floods due to unprecedented natural resource degradation. Machine learning systems have been used for predicting landslide risk of camp infrastructure from geospatial variables (Ahmed, Firoze, and Rahman 2020). Furthermore, early warning system for landslides in Rohingya camps and Cox's Bazar have been developed (Ahmed et al. 2020).

Authorities responsible for refugee management require automated methods for maintaining managerial efficiency at such a sheer scale. Gathering refugee population census data at regular intervals is challenging in developing countries like Bangladesh due to costs of field surveys and remoteness of the camps. Moreover, management and record keeping of refugee camp buildings are challenging because they are subject to constant change, i.e. new buildings are being constructed and previously existing buildings are being replaced by larger/newer buildings or being altered. This leads to inaccurate and noisy building maps when retrieved from OpenStreetMap vector data. The importance of reliable building maps come as an instrumental tool for broader applications in land use and land cover mapping, such as efficient urban planning and environmental impact assessment (Zhang et al. 2020). Furthermore, updated monitoring data of refugee camp buildings is important in the context of aid delivery since the relief community needs to have precise spatial information for dispensing essential aid to refugees (Bjorgo 2000). Remotely sensed data comes as a less labour-intensive means to deal with the aforementioned obstacles (Ahmed et al. 2019b; Srivastava, Vargas-Muñoz, and Tuia 2019).

The availability of ultra-high-resolution drone images collected by unmanned aerial vehicles (UAVs) coupled with building maps from OpenStreetMap, albeit being noisy, can enable the detection and localization of camp buildings in drone images. Aerial images, which include images collected by drones and aircraft, have higher spatial resolution than satellite images in most cases. In our case, the images were collected by drones (UAVs) and not by a low flying-piloted aircraft. Building extraction from remotely sensed data is a pixel labelling task, i.e. semantic segmentation (falling under the broader domain of image-to-image translation), where each pixel is labelled as either a building or a non-building. In recent times, deep learning has become the de facto solution for computer vision problems, extending its applicability to remote sensing and building extraction. Pixel level semantic segmentation can be used for generating camp building maps by learning to extract buildings from drone images of Rohingya refugee camps. The contributions of this work are as follows:

(i) The U-Net deep learning architecture is compared to the pixel classification tool of the most widely used Geographic Information System (GIS) software on the market.
(ii) We outline measures for restraining learning systems from overfitting to noisy labels, allowing reliable learning under noisy settings.
(iii) The effects of label noise on numeric performance metrics are quantified by comparing against noise-free labels.
(iv) Our system can learn to generate building maps which are less noisy and more accurate than freely available noisy labels from OpenStreetMap.
(v) Given the issues identified, nonconvex noise robust loss functions with dynamically learned hyperparameters are introduced as another potential solution to learning from noisy labels.
(vi) Spatio-temporal buildings maps of Kutupalong megacamp are generated and a wide variety of potential applications are indicated.

The remaining text is organized as follows. Section 2 describes related work on deep learning used for building extraction, indicating similarities and differences with existing work. Section 3 provides problem formulation followed by data description, and then goes on to describe learning systems for semantic segmentation, their training configurations and experimental setups. Section 4 contains experimental results, analysis and interpretations. Section 5 outlines potential applications of this work.

2. Deep learning for building extraction

Infrastructure mapping in the form of building and road extraction through deep learning has provided state of the art results consistently. Models developed for building extraction have been applied for road extraction and vice versa. For supervised learning, unavailability of sufficient labels and disadvantages of re-training/re-clustering techniques for labelling leads to noisy labels and latency, Hierarchical Guided Cluster Labelling (HGCL) produces less noisy labels with less effort when compared to aforementioned methods (Wigness, Draper, and Ross Beveridge 2015). The study aimed to minimize label noise during automated labelling. In our case, the data have been already labelled by humans (i.e. OpenStreetMap contributors) and consequently, noise has already been incorporated into the labels. We aim to learn less noisy representations from labels which are already available, albeit being noisy. Advent of the generalized delta rule enabled the training of artificial neural networks with hidden units for learning internal representations (Rumelhart, Hinton, and Williams 1985). Fully connected deep neural networks with hidden units, as conceptualized by Rumelhart, Hinton, and Williams (1985), have been applied for building extraction (Lari and Ebadi 2007) and road extraction (Mnih and Hinton 2010). Mnih and Hinton (2010) use an image patch size much larger than the predicted map patch size. Larger size of the input image patch (compared to the output map patch size) is beneficial since the network is provided more information in the form of context. The roof of a building and the centre of a road may appear identical to the network if the patch size is too small. Building maps can often be incomplete or be poorly registered, leading to noisy labels. Mnih and Hinton (2012) address omission noise (missing labels) and registration noise (shifted labels) by introducing noise robust loss functions.

Convolutional neural networks outperformed previously tested deep learning methods for building and road extraction (Saito and Aoki 2015). The convolution operation extracts spatial information by allowing the network to learn intermediate features with spatial semantics. Convolutional layers are usually followed by downsampling layers for learning larger and higher-order features in images. However, the detailed structure of the input is lost due to downsampling since it produces a coarser representation. For semantic segmentation, the structure of the input being represented in the output is as important as the predicted classes for pixels. When considering previously mentioned methods, prediction density is much lower than input density, i.e. input image patch size is larger than output map patch size. Sherrah (2016) applied the no-downsampling approach for the fully convolutional network, which had already been applied for semantic segmentation in Long, Shelhamer, and Darrell (2015), showing the memory advantages compared to patch-based approaches. This method ensured that prediction density is maximum, i.e. equal to input density, since no downsampling was used. Other approaches to the fully convolutional network did not opt to remove downsampling (Marmanis et al. 2016). Rather, the deconvolution operation was used for upsampling the data in later layers of the network. Skip connections from previous layers are concatenated with upsampled layers to architecturally prioritize both prediction density and classification. Yuan (2017) compares the base fully convolutional network to their network which uses bilinear interpolation for upsampling; finding the latter superior in performance.

The U-Net (Ronneberger, Fischer, and Brox 2015) builds on the idea of the fully convolutional network. The contracting path, often replaced by traditional convolutional neural networks for image classification, leads to the bottleneck of the network which then leads to the expanding path where upsampling layers produce finer representations to be concatenated with skip connections coming from layers of the contracting path. The entire network is architecturally symmetric about the bottleneck. The contracting and expanding paths are referred to as the encoder and decoder, respectively, and the entire U-Net can be thought of as an image-to-image translation autoencoder with skip connections for supervised semantic segmentation. The current state of the art models for building extraction are based on the U-Net. For example, the architecture has been applied for multi-class semantic segmentation of remotely sensed data (Kim et al. 2018). Incorporating residual blocks into the U-Net gave rise to the deep residual U-Net (Res-U-Net) which has been used along with guided filters for building extraction (Xu et al. 2018). Lu, Sun, and Ong (2018) proposed a modified U-Net for using two different spatial resolutions. Similarly, Ji, Wei, and Lu (2019) introduces a scale/spatial resolution robust model equipped to take both aerial and satellite images as input. Inception modules have also been used in variants of the U-Net (Zhang et al. 2019). As such, we employ the U-Net architecture for supervised semantic segmentation of drone images of Rohingya camps. The Postdam and Vaihingen datasets are used as benchmarks for building detection tasks. However, label noise in Rohingya camps is intrinsically different from those datasets (Postdam and Vaihingen are cities in a first world country with very up-to-date and accurate labels compared to noisy labels of Rohingya camps). It is also observed that almost all labels are rectangles and have internal angles equal or approximately equal to 90°. This is because, for OpenStreetMap contributors working on large-scale annotation/tracing, speed is an important factor. Labelling tools used to capture generic shape and

size but not finer details. For most applications, this is sufficient. However, that is not the case for semantic segmentation since label noise directly affects learning.

3. Materials and methods

3.1. Problem formulation

Given an image $\mathbf{S} \in \mathbb{R}^{n_h \times n_w \times n_d}$, which can be represented as a rank 3 tensor with height n_h, width n_w and depth n_d, there is a corresponding building map $\widetilde{\mathbf{M}} \in \mathbb{R}^{n_h \times n_w}$ as shown in Equation (1). For our case, $n_h = n_w = 512,\ n_d = 3$ (for Red, Green and Blue bands) and $\mathbf{S} \in \mathbb{R}^{512\times512\times3}$ (refer to Section 3.2 for further details). The image pixels are encoded as 8-bit unsigned integers and take 2^8 possible discrete values, i.e. the set of possible values for $\mathbf{S}_{i,j,k}$ is $S \in \{0, 1, 2, \ldots 255\}$ where i, j and k are indices for row, column and channel, respectively. The corresponding binarised building map has a single channel and same row and column dimensions as $\mathbf{S}$ i.e. $\widetilde{\mathbf{M}} \in \mathbb{R}^{512\times512}$, where each pixel is indexed by i and j.

$$\widetilde{\mathbf{M}}_{i,j} = \begin{cases} 1, \text{if building} \\ 0, \text{if non} - \text{building} \end{cases} \tag{1}$$

Therefore, we essentially model the probability distribution of the building map given the corresponding image, given in Equation (2), where all pixels of $\mathbf{S}$ play a role in determining the probability of a specific map pixel $\widetilde{\mathbf{M}}_{i,j}$ being a building. It is assumed that each $\mathrm{P}\left(\widetilde{\mathbf{M}}_{i,j}|\mathbf{S}\right)$ is a Bernoulli distribution (since the random variable in question takes binary values) and are conditionally independent. Similarly, the predicted map $\widehat{\mathbf{M}}$ has the same dimensions as $\widetilde{\mathbf{M}}$. $\widehat{\mathbf{M}}_{i,j}$ approximates the mean of the distribution $\mathrm{P}\left(\widetilde{\mathbf{M}}_{i,j}|\mathbf{S}\right)$.

$$\mathrm{P}\left(\widetilde{\mathbf{M}}|\mathbf{S}\right) = \prod_{i=1}^{n_h}\prod_{j=1}^{n_w} \mathrm{P}\left(\widetilde{\mathbf{M}}_{i,j}|\mathbf{S}\right) \tag{2}$$

3.2. Data

The study area, comprising of eight regions/camps, is located in the district of Cox's Bazar in the southeastern border region of Bangladesh (Figure 1). Three regions (Kutupalong North, Kutupalong South and Choukhali) fall under Ukhia upazila (sub-district) other five fall under Teknaf upazila (sub-districts).

The images were collected by the International Organization for Migration (IOM) for the Needs and Population Monitoring project. Acquisition dates of images of all eight regions used for training, validation and testing are within the same month (September 2018) (refer to the supplemental material for details). The images are of ultra-high-resolution, having spatial resolution of 10 cm, which is extremely detailed and requires large memory for storage.

The building footprint maps are obtained from OpenStreetMap vector data. The supplemental material shows vector statistics of the eight camps. Building polygons are labelled by hand by OpenStreetMap contributors. However, errors in labelling are often observed. These errors transform into label noise when binarized raster building maps are

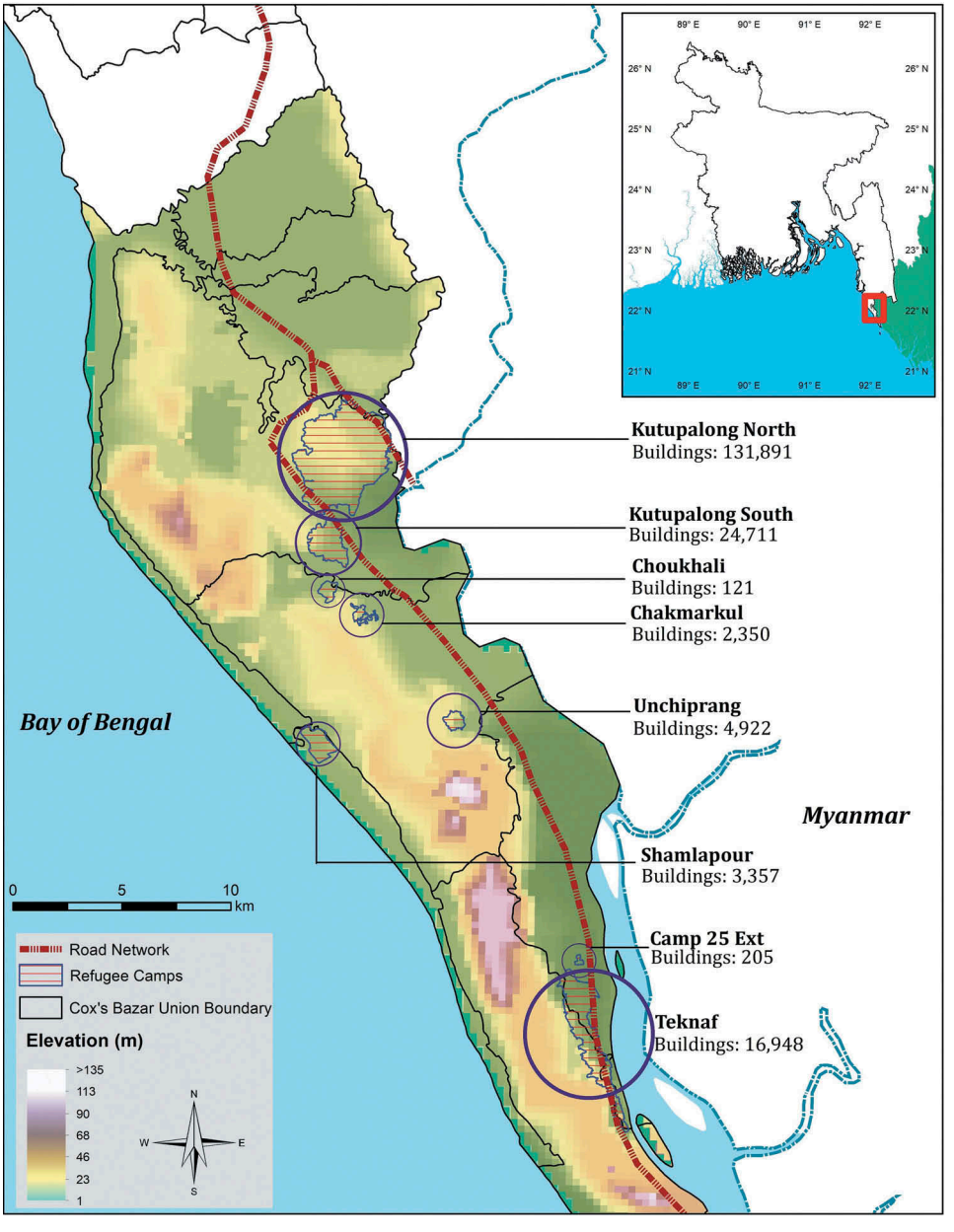


Figure 1. Map of study area shows locations of Rohingya refugee camps in southeastern border region of Bangladesh.

created from the shapefiles. Although such errors do not affect most general purpose uses of building footprint shapefiles, in the case of semantic segmentation, the labels are generated from these shapefiles and will have direct effects on training and testing. Mnih and Hinton (2012) recognized this issue as a source of reduction in performance. There are various types of label noise observed in the camp building shapefiles when compared against corresponding drone images (Figure 2).

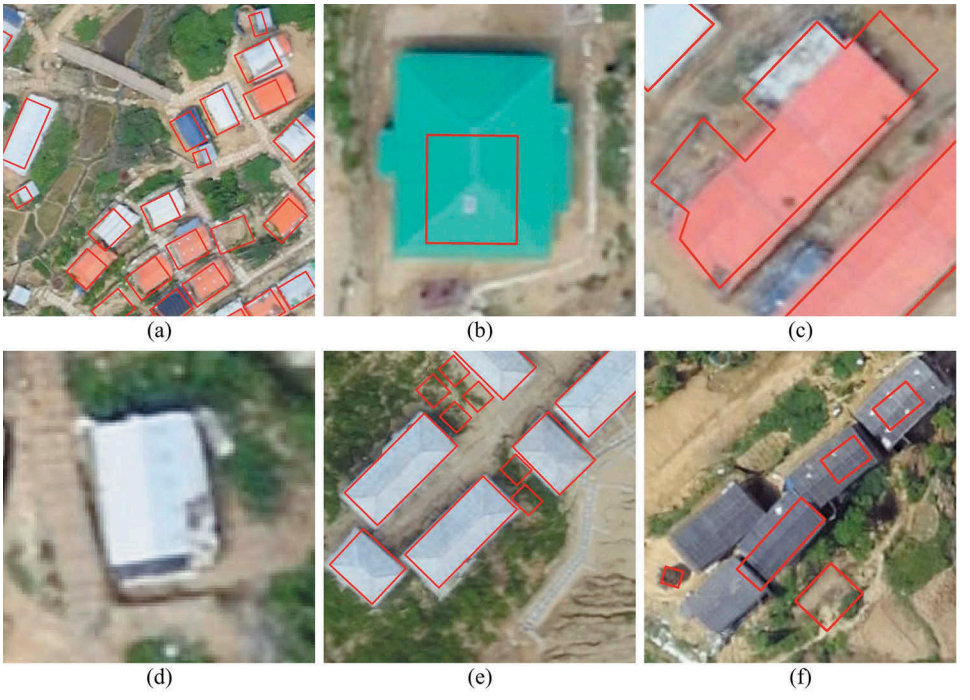


Figure 2. Examples of label noise. (a) label shift; (b) label shrinkage; (c) label enlargement; (d) label omission, (e) incorrect label presence; (f) combination.

One of the most commonly observed form of label noise is label shift (Figure 2(a)) where the dimensions and orientation of building shapefiles correspond to the image but is translated/shifted in some arbitrary direction. Label shrinkage (Figure 2(b)) and label enlargement (Figure 2(c)) refer to cases where the building labels are smaller than the buildings in images and larger than the buildings in images, respectively. Label omission (Figure 2(d)) occurs where buildings existing in images are missing in building shapefiles. This is because, parts of camp building maps are not always up-to-date since construction of new camp buildings is occurring at a rapid pace. The opposite form of noise is also observed where buildings labels exist but there are no buildings in the images (Figure 2(e)). A combination of label noises can also be observed (Figure 2(f)). Thus, we refer to $\widetilde{\mathbf{M}}_{i,j}$ as the noisy labels, and we use the notation $\mathbf{M}_{i,j}$ to refer to noise-free labels (Equation (3)). The label noise term ξ can flip a noise-free label (Equation (4)). These label noises cause the penalization of classifiers for making correct and confident predictions, underestimating classifier performance.

$$\widetilde{\mathbf{M}}_{i,j} = \mathbf{M}_{i,j} + \xi \tag{3}$$

$$\xi = \begin{cases} 1, \text{if } \mathbf{M}_{\mathrm{i,j}} = 0 \text{ and label is to be flipped} \\ -1, \text{if } \mathbf{M}_{\mathrm{i,j}} = 1 \text{ and label is to be flipped} \\ 0, \qquad \text{if no changes are to be made} \end{cases} \tag{4}$$

Large drone images are clipped using camp outline shapefiles provided by Inter Sector Coordination Group (ISCG). These boundaries separate refugee infrastructure from civilian infrastructure. The building polygons in vector shapefiles are rasterized using the corresponding input aerial image's metadata, i.e. spatial resolution, coordinate reference system, rows, columns, extent, latitudes, longitudes, etc., resulting in a raster building map where each pixel corresponds to a specific pixel in the drone image. The large drone images and rasterized buildings are tiled (cropped) to form 512×512-pixel tiles. Image tiles have three channels (RGB) whereas label maps have a single channel.

3.3. Learning to extract buildings

3.3.1. U-Net

The U-Net is a derivation of fully convolutional networks since it does not use any fully connected layers. The contracting path (as shown in Figure 3) uses downsampling blocks (D_1, D_2, D_3 and D_4 in Figure 3) where convolutional filters learn representations and are then downsampled using max-pooling and filter size of 2×2, halving the row and column size. Each block doubles the number of filters and halves height and width when compared to its previous block. For example, U_1 outputs tensors of dimensions $512 \times 512 \times 64$, U_2 outputs tensors of dimension $256 \times 256 \times 128$ and so on. The bottleneck (Figure 3) has 1024 filters where height and width have been reduced by a factor of 16 compared to input height and width. In convolutional neural networks for image classification, this bottleneck is usually connected to fully connected layer(s) to obtain classifications. However, the U-Net is designed for semantic segmentation. It uses upsampling blocks (U_1, U_2, U_3 and U_4 in Figure 3) to incorporate structural information into predictions. Outputs of downsampling blocks are copied and concatenated with inputs to upsampling blocks. Each upsampling block halves the number of filters and doubles height and width. This output is concatenated with output of downsampling blocks. All convolution layers use same padding. Height and width of tensors change only during upsampling and downsampling layers. The model has approximately 31 million trainable parameters. Output of convolutional layers is fed into leaky Rectified Linear Unit (ReLU) activation functions with $\alpha = 0.2$ i.e. the function outputs $\max(0.2 \times x, x)$ and final pixel-level predictions are obtained using the sigmoid activation function $1/(1 + \exp(-x))$ where x is the input to the activation functions.

3.3.2. Maximum likelihood estimation in ArcGIS

Maximum likelihood classification also known as discriminant analysis is a well-established pattern recognition approach that operates calculating multidimensional distances to classify clusters (Hogland, Billor, and Anderson 2013). Based on the decision rule of Bayes theorem, maximum likelihood accepts all classes as Gaussian distributions. It has been a popular tool for supervised classification in the ArcGIS platform (Strahler 1980; Foody et al. 1992; Dewan and Yamaguchi 2009; Bhuiyan et al. 2019; Islam et al. 2019). Comparison of maximum likelihood classification and artificial neural network has been made in the case of soil mapping (Chagas, Vieira, and Fernandes Filho 2013).

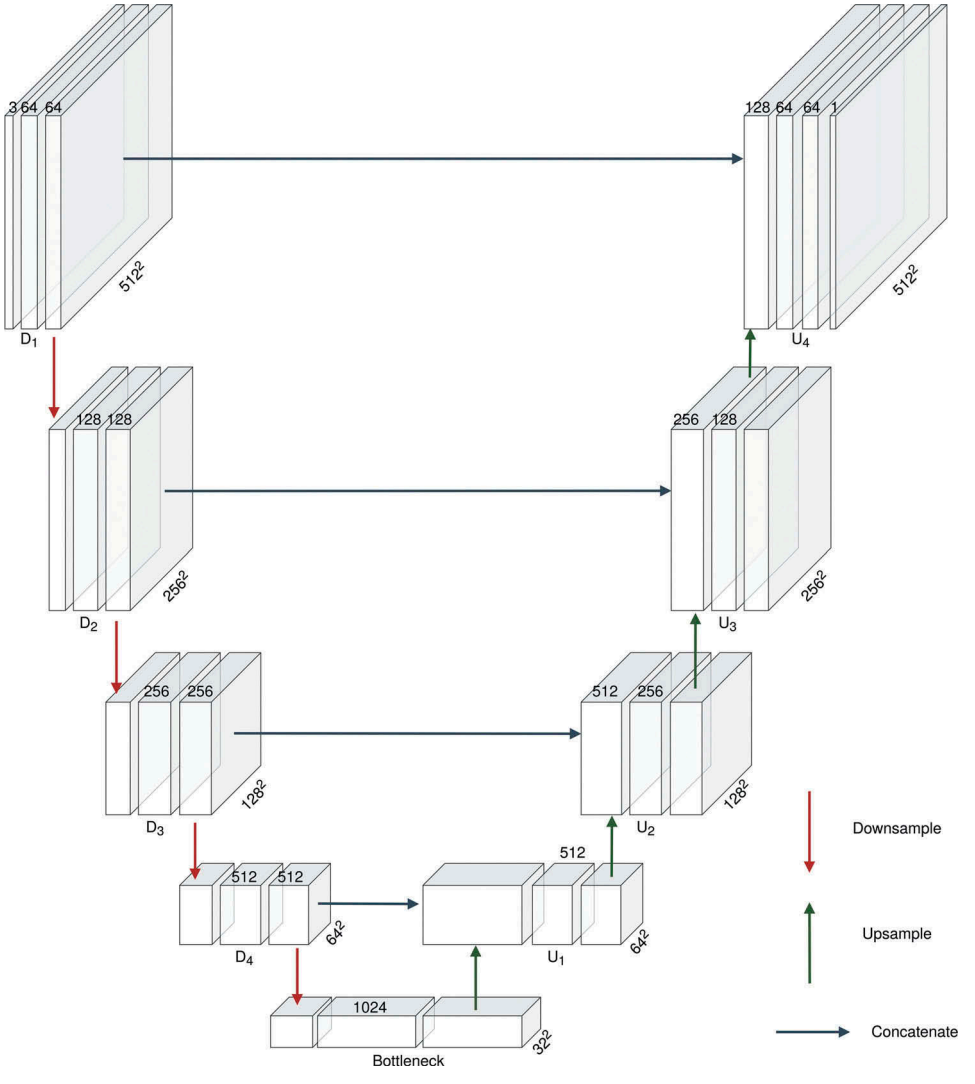


Figure 3. Architecture of the U-Net used in this study. Downsampling blocks (D_1, D_2, D_3 and D_4) learn higher-order features. Upsampling blocks (U_1, U_2, U_3 and U_4) and skip connections (concatenate) incorporate spatial information back into predictions.

3.4. Experimental setup

The binary cross entropy function (Equation (5)) determines the loss of the network at each iteration, and instructing, through its gradients, how well or poorly a network is performing with respect to the network's current weights and how they should be changed. The objective during training is to minimize this loss function with respect to the weights/parameters. Thus, any assumptions made by the loss function will back propagate throughout the weight change step. Noisy labels are not considered in these settings and all labels are assumed to be noise free. These assumptions may not hold in cases where noisy labels are prevalent and cause the network to be unnecessarily penalized.

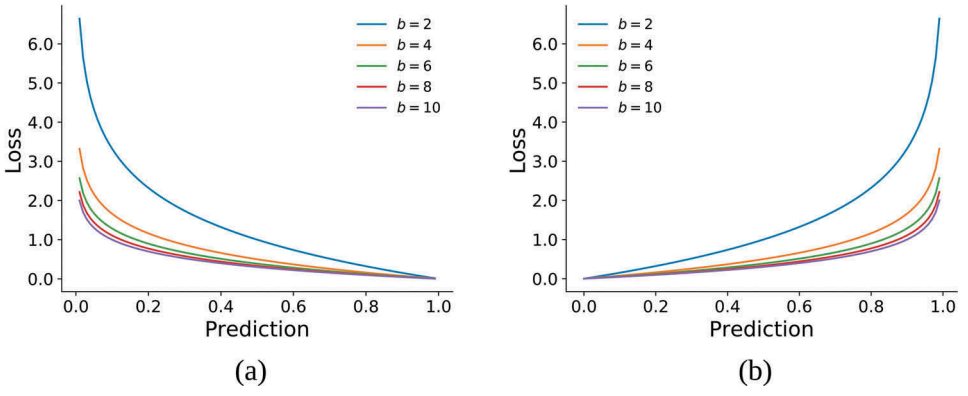


Figure 4. Binary cross-entropy loss for (a) $\widetilde{\mathbf{M}}_{i,j} = 1$ and (b) $\widetilde{\mathbf{M}}_{i,j} = 0$. Prediction refers to $\widehat{\mathbf{M}}_{i,j}$.

$$-\frac{1}{n_h \times n_w}\sum_{i=1}^{n_h}\sum_{j=1}^{n_w}\left[\widetilde{\mathbf{M}}_{i,j}\,\log_b\widehat{\mathbf{M}}_{i,j} + (1-\widetilde{\mathbf{M}}_{i,j})\log_b\left(1-\widehat{\mathbf{M}}_{i,j}\right)\right] \tag{5}$$

The loss depends on the negative log likelihood of the predictions (Figure 4). In the presence of label omission noise when noisy label $\widetilde{\mathbf{M}}_{i,j} = 1$ (Figure 4(a)) and noise-free label $\mathbf{M}_{i,j} = 0$ the correct prediction $\widehat{\mathbf{M}}_{i,j} \approx 0$ is penalized by high loss. The prediction and loss in Figure 4(a) refer to $\widehat{\mathbf{M}}_{i,j}$ and $(\widetilde{\mathbf{M}}_{i,j}\,\log_b\widehat{\mathbf{M}}_{i,j})$ respectively. Similarly, in the presence of incorrect presence noise when noisy label $\widetilde{\mathbf{M}}_{i,j} = 0$ (Figure 4(b)) and noise-free label $\mathbf{M}_{i,j} = 1$ the prediction $\widehat{\mathbf{M}}_{i,j} \approx 1$ is penalized by high loss as well. The prediction and loss in Figure 4(b) refer to $\widehat{\mathbf{M}}_{i,j}$ and $\left(\left(1-\widetilde{\mathbf{M}}_{i,j}\right)\log_b\left(1-\widehat{\mathbf{M}}_{i,j}\right)\right)$ respectively.

The U-Net is trained on both the standard training set (Figure 5) as well as an augmented training set which includes rotated, colour varied, translated, sheared versions of tiles in the standard training set increasing the number of tiles by a factor of 7. We refer to this version of the model as the U-Net-AUG. The U-Net-AUG model is trained on over 100 gigabytes of augmented data. This allows generalization to new images, even under noisy settings. Extensive dropout in the contracting path reduces overfitting to noisy labels. We use the Adam optimizer (Kingma and Ba 2014) with a learning rate of 10^{-4}. The model is trained for 5 epochs. As specified by Ronneberger, Fischer, and Brox (2015), we also opt to use a batch size of 1 because larger input size is favoured over larger batch sizes. The maximum likelihood estimation model in image classification tool of the ArcGIS platform was trained using a signature file generated from vector shapefile of buildings.

The model is tested on the southern part of Kutupalong North (Figure 5), since it is the largest camp which includes 71% of all camp buildings. At tile size of 512 × 512 pixels, the models are tested on approximately 4000 image tiles. Excluding empty tiles, such as those on the ragged camp borders (Figure 5), results in approximately 2500 tiles. Training set includes northern part of Kutupalong North and other regions. The U-Net model has been implemented in TensorFlow 2.0.

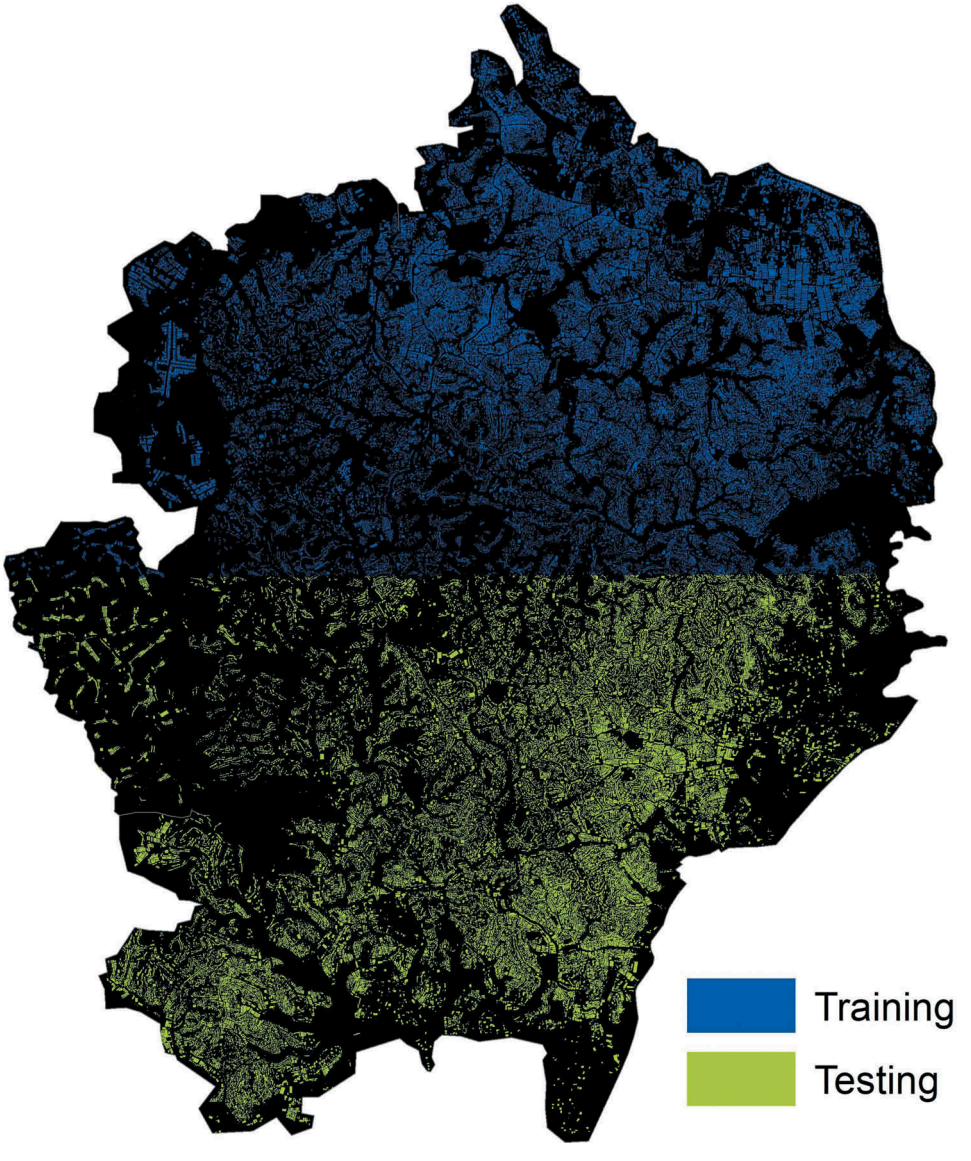


Figure 5. Kutupalong mega-camp (Kutupalong North) is split in half for delineating training and testing data. 131,891 buildings are shown.

4. Results and discussion

4.1. Quantitative evaluation of performance

Classifier performance is evaluated using several metrics (Equation (6) and Table 1). True positives (TP), true negatives (TN), false positives (FP) and false negatives (FP) from testing data are used for calculations. The Accuracy (*A*) (Equation (6)) gives an overall indication of performance. However, we test using other metrics due to class imbalance, e.g. Precision (*P*), Recall (*R*), Dice coefficient (DC) (Sørensen–Dice coefficient) (also commonly known as the F1-score) and Intersection-over-Union (IoU) (also known as the Jaccard index).

Table 1. Formulae for calculating different performance metrics with respect to the two classes, i.e. building and non-building.

Metric	Non-building ($M_{i,j} = 0$)	Building ($M_{i,j} = 1$)
Precision	$P = \frac{(TN)}{(TN)+(FN)}$	$P = \frac{(TP)}{(TP)+(FP)}$
Recall	$R = \frac{(TN)}{(TN)+(FP)}$	$R = \frac{(TP)}{(TP)+(FN)}$
Dice coefficient	$DC = \frac{2(TN)}{2(TN)+(FN)+(FP)}$	$DC = \frac{2(TP)}{2(TP)+(FP)+(FN)}$
Intersection-over-Union (IoU)	$IoU = \frac{(TN)}{(TN)+(FN)+(FP)}$	$IoU = \frac{(TP)}{(TP)+(FP)+(FN)}$

Considering the set of predictions and the set of labels, the IoU is the ratio between the number of intersecting pixels to the number of pixels in the union of both sets.

$$A = \frac{(TP) + (TN)}{(TP) + (TN) + (FP) + (FN)} \times 100 \tag{6}$$

4.1.1. Evaluating noisy settings

The Maximum Likelihood Estimation (MLE) model, U-Net, and U-Net-AUG (U-Net trained on augmented dataset) are tested on the entire southern half of Kutupalong North mega-camp (approximately 4000 images). However, a large proportion of these images are empty because they are located in close proximity to rectangular extent of the un-tiled large image and at the ragged outline of the camp region. In order to test more realistic settings, we also separately test the three models solely on images which are not empty. There are six different configurations which are tested, as shown in Table 2. The first three configurations in Table 2 are tested on entire test set and the last three configurations are tested on filtered test set (suffixed by F in Table 2) which exclude empty images. Even though performance metrics with respect to both classes have been calculated, metrics for class building ($\mathbf{M}_{i,j} = 1$) are of special interest.

Even though accuracy of all models is relatively high, judging performances solely on accuracy are not ideal because there is massive class imbalance in images. The number of non-building pixels is approximately 83% of the total camp area and number of pixels labelled as buildings is approximately 17% of the total camp. This calculation excludes non-building pixels in area between raster extent rectangle and camp outline, the inclusion of which would further reduce the building to non-building pixel ratio. In such cases, high accuracy is mostly due to high true negative rate even though true positive rate may be low. Even then, U-Net configurations have higher accuracy than corresponding MLE

Table 2. Performance of different classifiers (MLE, U-Net and U-Net-AUG) on standard test set (non-suffixed) and filtered test set (suffixed by -F).

		Non-building ($M_{i,j} = 0$)				Building ($M_{i,j} = 1$)			
Method	Accuracy (%)	Precision	Recall	Dice coefficient	IoU	Precision	Recall	Dice coefficient	IoU
MLE	80.02	0.97355	0.79744	0.87674	0.78052	0.33163	0.82267	0.47271	0.30951
U-Net	91.41	0.95694	0.94607	0.95148	0.90744	0.59935	0.65457	0.62574	0.45533
U-Net-AUG	91.81	0.95823	0.94944	0.95382	0.91171	0.61818	0.66418	0.64035	0.47097
MLE-F	67.60	0.93902	0.64157	0.76231	0.61591	0.35031	0.82267	0.49138	0.32572
U-Net-F	85.14	0.91626	0.89814	0.90711	0.83001	0.60429	0.65457	0.62842	0.45817
U-Net-AUG-F	85.82	0.91891	0.90437	0.91158	0.83753	0.62270	0.66418	0.64277	0.47359

configurations. Precision, recall, Dice coefficient and IoU scores reflect differences in performance due to filtering.

The precision of MLE configurations is much lower than U-Net configurations because false-positive rate is extremely high (Figure 7). Non-building pixels are labelled as buildings by the MLE models resulting in very noisy predictions albeit having high true positive rate which attributes for high recall (Table 2 and Figure 7). However, overall Dice coefficient and IoU of MLE are lower compared to U-Net configurations. Figure 7 also shows that numeric performance metrics (shown in Table 2) underestimate the actual performance due to label noise because these metrics are calculated using noisy labels whereas the predictions, even though being correct, are penalized for being considered incorrect due to said noisy labels.

4.1.2. Evaluating noise-free settings

As mentioned previously, the performance metrics shown in Table 2 were calculated against noisy labels. In order to gauge true performance, we relabel a subset of the testing data by hand for removing noise. Labelling the entire test set of 2500 images (655 million pixels) would take 10,500 hours (438 days of non-stop labelling) which is not realistically feasible. A subset of 22 million pixels (84 large tiles) is relabelled which has taken approximately 20 h to do by hand for obtaining noise-free labels. That averages to 1.1 million pixels (4.2 images) per hour. We refer to this relabelled dataset as NOISE-FREE. The corresponding noisy labels for that subset of testing data from OpenStreetMap is referred to as NOISY. First, we calculate the similarity between NOISY and NOISE-FREE. In this scenario, NOISY is treated as the output of a hypothetical classifier and NOISE-FREE is used as the ground truth (first row in Table 3). We then test the predictions made by U-Net and U-Net-AUG against NOISY and NOISE-FREE. As shown in Table 3, same model, whether it be U-Net or U-Net-AUG, shows drastically different performance when comparing against NOISY and NOISE-FREE. It is apparent that performance is grossly underestimated when calculated against NOISY, since the same predictions when evaluated against NOISE-FREE, indicate much better performance. Testing against NOISE-FREE reveals these negative effects of noisy labels. The IoU score gain of U-Net on NOISE-FREE over U-Net on NOISY is approximately 19% whereas the IoU score gain of U-Net-AUG on NOISE-FREE over U-Net-AUG on NOISY is approximately 21%. This confirms that deriving performance metrics against noisy labels greatly underestimates true classifier

Table 3. Performance of U-Net and U-Net-AUG on NOISY and NOISE-FREE test sets. Performance of NOISY being treated as output of a hypothetical classifier evaluated against NOISE-FREE is also shown.

		Non-building ($M_{i,j} = 0$)				Building ($M_{i,j} = 1$)			
Method	Accuracy (%)	Precision	Recall	Dice coefficient	IoU	Precision	Recall	Dice coefficient	IoU
NOISY on NOISE-FREE	90.431	0.92375	0.96420	0.94354	0.89312	0.77901	0.61329	0.68629	0.52240
U-Net on NOISY	90.445	0.94520	0.94437	0.94478	0.89534	0.64361	0.64726	0.64543	0.47648
U-Net-AUG on NOISY	90.910	0.94531	0.94995	0.94763	0.90046	0.66704	0.64592	0.65631	0.48844
U-Net on NOISE-FREE	93.953	0.94449	0.98497	0.96431	0.93108	0.90778	0.71872	0.80226	0.66982
U-Net-AUG on NOISE-FREE	94.657	0.94597	0.99224	0.96855	0.93903	0.95054	0.72464	0.82236	0.69831

performance. Data augmentation creates artificial samples from existing data. For our case, the superior performance of U-Net-AUG in all cases indicates that the augmented data provide added discriminative information to classifiers when compared to the standard un-augmented training set. For example, rotated and flipped samples in augmented training set compensate for the absence of samples with wide variety of building configurations/alignments in the un-augmented training set. This helps the classifier in generalizing better to previously unseen configurations/alignments of buildings, resulting in better testing performance. Similar case can also be observed for augmented images with varying levels of zooming which makes the U-Net-AUG more scale-robust by being able to classify buildings with wider range of sizes. The ratio of mislabelled pixels to correctly labelled pixels in the standard training set is higher than the ratio of mislabelled pixels to correctly labelled pixels in the augmented training set.

Better performance on NOISE-FREE compared to performance on NOISY shows that the classifiers learn less noisy representations which are reflected by the predicted testing labels being more similar to NOISE-FREE than to NOISY When comparing the performance metrics of NOISY against NOISE-FREE and the U-Net configurations against NOISE-FREE, the latter has superior performance with regard to every single performance metric. Therefore, the U-Net and U-Net-AUG model are able to generate more accurate and less noisy labels than freely available building maps from OpenStreetMap. Labels generated by our system provide IoU gain of 17.6% and Dice score gain of 13.6%. Relabelling noisy testing data allowed the testing of such configurations, revealing the dynamic effects of label noise on numeric performance metrics. Data augmentation reduces overfitting and enables learning from noisy labels. Similarly, extensive use of dropout in the downsampling blocks of the contracting path also reduces overfitting. U-Net-AUG on NOISE-FREE provides IoU and Dice score gain of 2.3% and 2%, respectively, over U-Net on NOISE-FREE for the building class. Furthermore, U-NET-AUG on NOISE-FREE also provides increased precision (4.3% gain) and recall (0.6% gain) when compared to U-Net on NOISE-FREE. Thus, the classifier trained on massively augmented data learn less label noise.

4.2. Qualitative evaluation

The feature maps learned are visualized and shown in Figure 6. Each column of feature maps corresponds to the output of a specific block of the network. For brevity, only the first four feature maps of blocks' outputs are shown. In the contracting path, as the tensors go towards the bottleneck, spatial information is lost due to downsampling layers which is incorporated back into the network using skip connections which copy and concatenate outputs of downsampling blocks (Figure 6). Feature maps of the bottleneck have learned more discriminative representations compared to feature maps of U1 whereas feature maps of U4 has much finer structural information incorporated.

Example of results shown in Figure 7 reveals that label omission noise (Figure 7 top row) and combined label noise (Figure 7 second row) is handled by the models. The same case is observed for and incorrect label presence noise (Figure 7 third row) which are also handled. It is important to note that models make correct predictions at the cost of numeric performance metrics. However, in some cases the model fails to provide consistent results. Almost all buildings in Rohingya camps are single-storied structures with

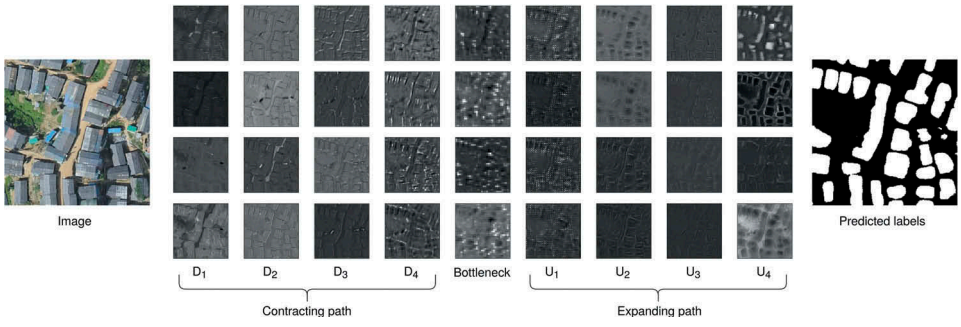


Figure 6. Feature maps learned from ultra-high-resolution drone images. The first four feature maps from the output of each block have been shown.

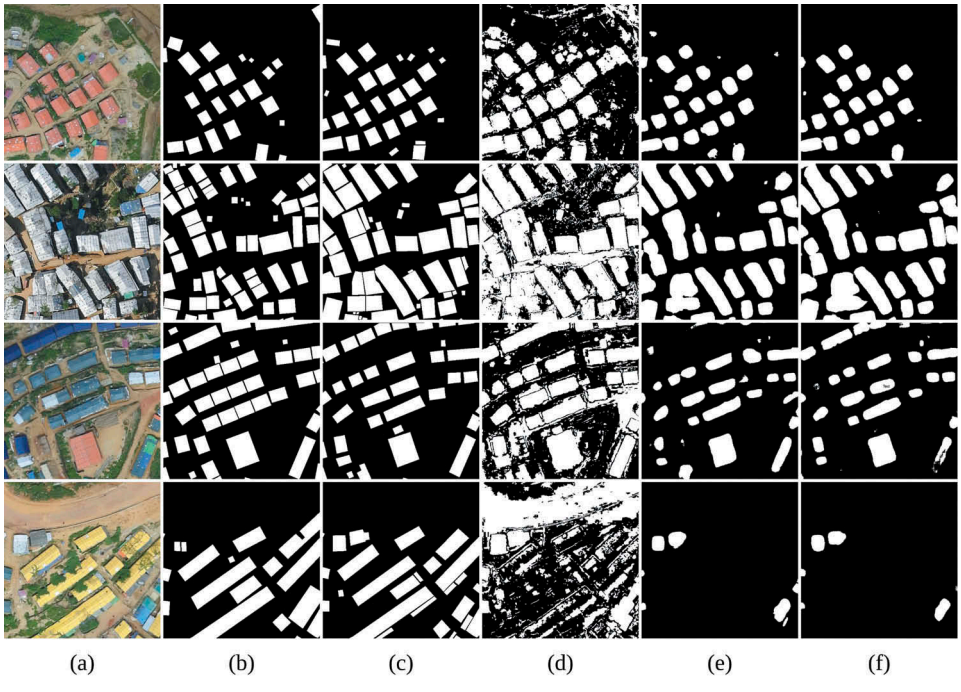


Figure 7. Example of results on the test set of southern Kutupalong megacamp (Kutupalong North). (a) Input image; (b) Noisy labels; (c) Noise-free labels; (d) Maximum Likelihood Estimation (ArcGIS); (e) U-Net; (f) U-Net-AUG.

corrugated metal roofs; most of which are un-coloured/unpainted. Some of these camp buildings have painted roofs or coloured corrugated plastic roofs (Figure 7 bottom row).

Rare colours of roofs can throw off the models and be undetectable, especially if training set does not contain similar samples. Models are successfully able to detect the few buildings with un-coloured roofs but majority of the buildings in the image with yellow roofs remain undetected. Label shrinkage noise can cause the models to lose structural integrity of predictions is some cases. Therefore, when there is label noise in the training and testing raster patches, the classifiers' actual performance can be much better than otherwise indicated by performance metrics.

4.3. Noise robust loss functions

Robust loss functions which factor in label noise can come to aid in the presence of noise in data. The binary cross-entropy loss is convex and penalizes the model for making confident and correct predictions. Nonconvex losses can alter the behaviour of the loss function (based on certain hyperparameters) to penalize confident predictions less. For example, probability density function of the exponentially modified Gaussian (exGaussian) distribution can be used as components of a noise-robust loss function (Equations (7) and (8))[1]. As Figure 8 shows, the loss is reduced if the model is confident enough about a prediction. Thus, the model is penalized less for making confident predictions. The prediction and loss in Figure 8(a) refer to $\widehat{\mathbf{M}}_{i,j}$ and $\left(\widetilde{\mathbf{M}}_{i,j}\exp\left(\frac{\lambda}{2}\left(2\mu+\lambda\sigma^2-2\widehat{\mathbf{M}}_{i,j}\right)\right)\operatorname{erfc}\left(\frac{\mu+\lambda\sigma^2-\widehat{\mathbf{M}}_{i,j}}{\sqrt{2}\sigma}\right)\right)$ respectively. The prediction and loss in Figure 8(b) refer to $\widehat{\mathbf{M}}_{i,j}$ and $\left(\left(1-\widetilde{\mathbf{M}}_{i,j}\right)\exp\left(\frac{\lambda}{2}\left(2\mu+\lambda\sigma^2-2\left(1-\widehat{\mathbf{M}}_{i,j}\right)\right)\right)\operatorname{erfc}\left(\frac{\mu+\lambda\sigma^2-(1-\widehat{\mathbf{M}}_{i,j})}{\sqrt{2}\sigma}\right)\right)$ respectively. Loss functions must be continuous and differentiable in the input space. Furthermore, automatic differentiation of newly developed loss functions must also be possible to maintain computational feasibility. Exploding and vanishing gradients can also limit the use of such functions.

$$-\frac{1}{n_h\times n_w}\sum_{i=1}^{n_h}\sum_{j=1}^{n_w}\left[\widetilde{\mathbf{M}}_{i,j}\exp\left(\frac{\lambda}{2}\left(2\mu+\lambda\sigma^2-2\widehat{\mathbf{M}}_{i,j}\right)\right)\operatorname{erfc}\left(\frac{\mu+\lambda\sigma^2-\widehat{\mathbf{M}}_{i,j}}{\sqrt{2}\sigma}\right)\right.$$
$$\left.+\left(1-\widetilde{\mathbf{M}}_{i,j}\right)\exp\left(\frac{\lambda}{2}\left(2\mu+\lambda\sigma^2-2\left(1-\widehat{\mathbf{M}}_{i,j}\right)\right)\right)\operatorname{erfc}\left(\frac{\mu+\lambda\sigma^2-\left(1-\widehat{\mathbf{M}}_{i,j}\right)}{\sqrt{2}\sigma}\right)\right] \tag{7}$$

$$\operatorname{erfc}(x)=\ 1-\frac{1}{\sqrt{\pi}}\int_{-x}^{x}\exp\left(-t^2\right)\mathrm{d}t \tag{8}$$

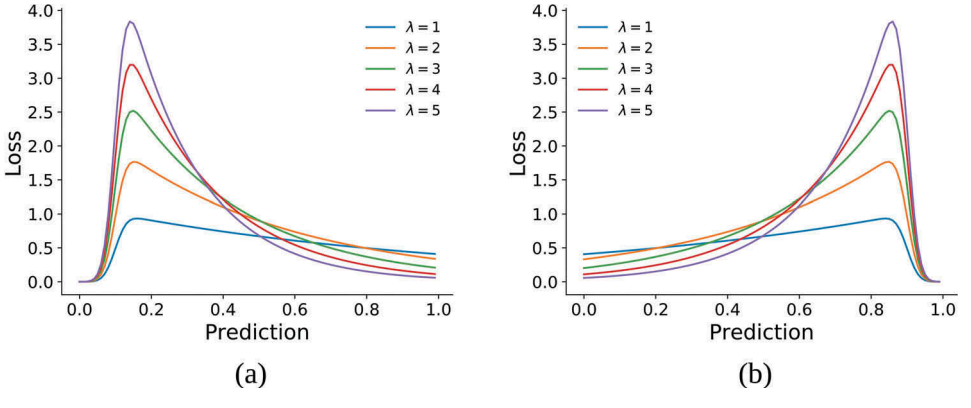


Figure 8. Noise robust binary loss function with probability density function of the exponentially modified Gaussian distribution as components.[2] (a) $\widetilde{\mathbf{M}}_{i,j}=1$ and (b) $\widetilde{\mathbf{M}}_{i,j}=0$. Prediction refers to $\widehat{\mathbf{M}}_{i,j}$.

4.4. Generating spatio-temporal building maps

Since the trained model is able to generate less noisy and more accurate labels, we segment drone images of Kutupalong North acquired on February 2018 and September 2018 to show the building increase over 7 months (Figure 9). The supplemental material lists all available drone images which can be used for generating building maps and outlines best practices. Since the model was never trained on images of February 2018 (because there are no temporally corresponding building maps), the predictions lose some structural integrity around building edges albeit overall outlines can be photointerpreted to be accurate. Large images acquired over different seasons have varying brightness, contrast, azimuth, quality, cloud cover, cloud shadow, etc. For best results, a subset of image tiles should be relabelled and used for fine tuning the trained model using transfer learning.

5. Applications

The models trained for the purpose of this study have a wide variety of applications in the context of refugee management and resource conservation. Spatio-temporal building maps can be used for locating regions where building construction/change has occurred, as well as track building density change in sub-camps along with a variety of other applications.

- *Calculating material stocks* of buildings will require accurate buildings maps, which can contribute to developing more realistic policies for lessening environmental burden generated from construction materials. Furthermore, current building material stock information will act as a benchmark for future materials flow studies and the prospects of circular economy (Heeren and Hellweg 2019). Construction materials for buildings generate greenhouse gases (GHGs), contributing to global climate change (Liu et al. 2019). In the light of resource decoupling, the current building stocks are now considered as a priority zone for urban mining since highly concentrated building clusters accommodate vast quantity of materials (De Tudela, Rose, and Stegemann 2020).
- *Net primary production* calculation can be aided by spatial information of anthropogenic structures such as building maps. Mahbub et al. (2019) calculated the human appropriation of net primary production (HANPP) in Bangladesh. However, urban area was not accounted for due to unavailability of accurate urban land cover maps. Building maps can assist region specific calculations of this kind. Human appropriation of net primary productivity (HANPP) is defined as an index to measure the health of a land use type due to anthropogenic disturbances. Higher value of HANPP indicates higher extent of land degradation for particular land use types.
- *Population mapping* can be performed using the drone images and predicted label maps as inputs to the models for population estimation. Census of Refugee populations is performed at much more regular intervals compared to national censuses. Population census through on-site field surveys can be slow and expensive. Building maps can aid building level population mapping of refugee camps.
- *Poverty mapping* has been performed using night-time satellite images (Jean et al. 2016). There is a semantic relationship to the structure and intensities of night-time

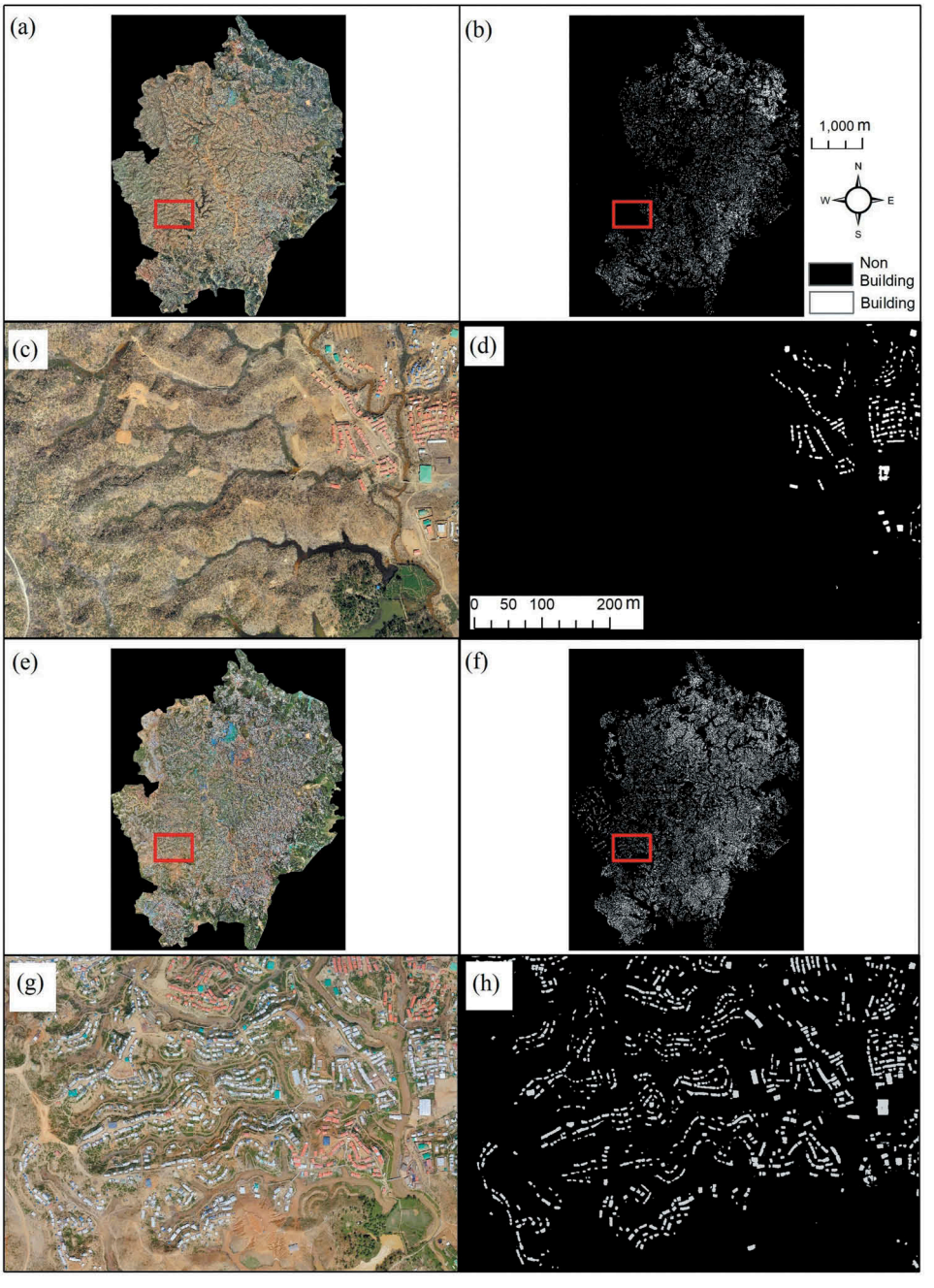


Figure 9. Spatio-temporal building maps of Kutupalong megacamp. (a) Image of 13 February 2018; (b) Building map of 13 February 2018; (c) Zoomed in view of highlighted region in (a); (d) Zoomed in view of highlighted region in (b); (e) Image of 24 September 2018; (f) Building map of 24 September 2018; (g) Zoomed in view of highlighted region in (e); (h) Zoomed in view of highlighted region in (f).

anthropogenic sources of light and economic activity. Binary building maps can provide information at a higher level of detail since night-time satellite imagery have coarser resolution.

- *Landslide and flood risk prediction* for buildings and infrastructure can have massive improvements with the aid of spatiotemporal building maps. Accurate building maps can serve as features for landslide risk prediction models along with traditionally used geospatial features such as elevation, Normalized Difference Vegetation Index (NDVI), Topographic Wetness Index (TWI) etc.
- *Environmental impact assessment* of Rohingya camps will also require accurate building maps for calculating the environmental footprints of these refugee camps. De Vries et al. (2017) performed an investigative environmental impact assessment of Kutupalong megacamp. Building maps will provide insights in calculating loss of natural resources through economic evaluation of environmental resources or goods.

6. Conclusions

This study provides a general framework for learning building extraction from ultra-high-resolution drone images and noisy labels from OpenStreetMap, especially for built and semi-built environments which are subject to constant change. Building maps of dynamic built environments are corrupted by label noise. Learning building detection from ultra-high-resolution drone images and noisy labels is possible. As the results have indicated that the U-Net can provide less noisy labels compared to the ArcGIS software classifier without any manual hand-engineered postprocessing, Label noise leads to underestimation of performance. Twenty-two million pixels are relabelled by hand to obtain noise-free labels as a subset of testing data. Testing noisy and noise-free labels allow calculation of effects of label noise. Our deep learning system achieves IoU gain of 17.6% and Dice score gain of 13.6% over freely available noisy labels from OpenStreetMap. Furthermore, our results outperform noisy labels in terms of every single performance metric considered.

Robust loss functions are a viable avenue for learning building extraction under noisy settings. Since noise distribution varies dataset to dataset, dynamic loss functions can cater to specialized noise distributions. The hyperparameters can be learned and updated throughout the training. Another interesting area to explore would be learning to predict completely noise-free labels from aerial images and noisy labels. This task would require large-scale noise-free data which would be very time consuming to obtain. Future work can also focus on using generative models for generating less noisy and more useful augmented data. This can especially aid learning from noisy labels if probability distributions of noise-free labels can be approximated for sampling noise-free augmented data.

Notes

1. Height and width of input images are denoted by n_h and n_w respectively. The hyperparameters λ, μ and σ determine the shape and positioning of the loss function and can be dynamically learned during training.
2. Hyperparameter values of $\mu = 0.1$ and $\sigma = 0.025$ are used.

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